Exercise 19.3

1. A force of 12 N acts for 5 s on a mass of 2 kg. What is the change in momentum of the mass ? What would be the change in momentum of a mass of 10 kg under the same condition ?

Sol^n: force act (F) = 12 N

time taken (t) = 5 secs.

change in momentum = ?

Case I : When mass (m) = 2 kg

Then $F = m \times a$ $\Rightarrow 12 = 2 \times a$ $\therefore a = 6\ ms^{-2}$

Again, $v = u + at$ $\Rightarrow v = 0 + 6 \times 5$ $\therefore v = 30\ ms^{-1}$

$\therefore$ Change in momentum = $mv - mu = 2 \times 30 - 2 \times 0 = 60\ kg\ ms^{-1}$

Case II : When mass (m) = 10 kg

$\therefore$ Change in momentum = $10(30 - 0)\ kg\ ms^{-1} = 300\ kg\ ms^{-1}$.

2. A body of mass 0.5 kg and initially at rest, is subjected to a force of 2 newtons for 1 sec. Calculate the change in momentum of the body.

Sol^n: mass of the body (m) = 0.5 kg

initial velocity (u) = 0

force exerted (F) = 2N

time taken (t) = 1 sec.

Then, $F = m \times a$ $\Rightarrow 2 = 0.5 \times a$ $\therefore a = 4\ ms^{-2}$

Again, $v = u + at$ $\Rightarrow 0 + 4 \times 1$ $\therefore v = 4\ ms^{-1}$

$\therefore$ Change in momentum = $mv - mu = m(v - u) = 0.5(4 - 0) = 2$ Ns.

3. A bullet of mass 0.006 kg travelling at 120 ms^{-1} penetrates deeply into a fixed target, and is then brought to rest in 0.01 s. Find

a) Change in momentum of the bullet

Sol^n: mass of the bullet (m) = 0.006 kg

velocity of the bullet (u) = 120 ms^{-1}

time taken (t) = 0.01 sec.

final velocity (v) = 0 m/s

$\therefore$ Change in momentum = $mv - mu = m(v - u) = 0.006 \times (120 - 0) = 0.72$ Ns.

b) The average retarding force exerted on the bullet

Sol^n: If F is the average retarding force exerted on the bullet

then, $F = \frac{\text{change in momentum}}{\text{time taken}} = \frac{0.72}{0.01}\ N = 72\ N.$

c) The distance of penetration of the target.

Sol^n: If a is the retardation of the system

Then $F = m \times a$ $\Rightarrow a = \frac{F}{m} = \frac{72}{0.006} = 12000\ ms^{-2}$

If s is the required distance of penetration of the target

Then $v^2 = u^2 - 2as$ $\Rightarrow 0 = 120^2 - 2.1200.s$ $\Rightarrow s = \frac{120^2}{2400} = 0.6$ m.

4. An inflated balloon contains 2.0 g of air which is allowed to escape from a nozzle at a speed of 4.0 ms^{-1}. Assuming that the balloon deflates at a steady rate in 2.5 s, what is the force exerted on the balloon ?

Sol^n: mass of the air (m) = 2 gm = 0.002 kg

initial speed of air (u) = 4 ms^{-1}

final velocity of air (v) = 0

time of duration (t) = 2.5 sec

Then F = mass × acceleration = mass × $\frac{\text{change in velocity}}{\text{time taken}}$

$= 0.002 \times \frac{4}{2.5} = 0.0032$ N.

5. A bullet of mass 10 g is fired from a rifle of mass 1000 g with a velocity of 50 kmh^{-1}. Find the velocity of the recoil of the rifle.

Sol^n: mass of the bullet (m) = $10g = \frac{10}{1000}$ kg = 0.01 kg

mass of the rifle (M) = 1000 g = 1 kg

muzzle velocity of the bullet (m) = 50 kmh^{-1}

recoil velocity of the rifle (V) = ?

Using the principle of conservation of linear momentum we have,

$mv = MV$

$\Rightarrow \quad 0.01 \times 50 = 1 \times V \qquad \therefore v = 0.5\ kmh^{-1}$.

6. A ball of mass 0.1 kg moving with a velocity of 6 ms^{-1}, collides directly with a ball of mass 0.2 kg at rest. Calculate their common velocity if both balls move together in the same direction.

Sol^n: Here, $m_1 = 0.1$ kg, $m_2 = 0.2$ kg, $u_1 = 6$ m/s $u_2 = 0$ m/s

Combined mass of balls = $(m_1 + m_2) = (0.1 + 0.2)$ kg

Let V be the velocity of the combination. Then

Using the principle of conservation of linear momentum we have,

$m_1 u_1 + m_2 u_2 = (m_1 + m_2) V$

$\Rightarrow \quad 0.1 \times 6 + 0 = (0.1 + 0.2) V \Rightarrow 0.3 V = 0.6 \qquad \Rightarrow V = 2\ ms^{-1}$.

7. Two bodies of masses 8 and 4 kg move along the x-axis in opposite directions with velocities 11 ms^{-1} and -7 ms^{-1} respectively. They collide and stick together. Find their velocity just after collision.

Sol^n: Here, $m_1 = 8$ kg, $m_2 = 4$ kg

$u_1 = 11$ m/s $u_2 = -7$ m/s

Let V be the velocity of the combination. Then using principle of conservation of linear momentum, the total momentum of the system remains constant. Hence momentum before collision = momentum after collision.

$m_1 u_1 + m_2 u_2 = (m_1 + m_2) V$

$\Rightarrow \quad 8 \times 11 + 4 \times (-7) = (8 + 4) V$

$\Rightarrow \quad 12V = 88 - 28 \qquad \therefore V = \frac{60}{12} = 5 \text{ ms}^{-1}.$

8. A 10 g bullet is fired from a kilogram gun suspended to move freely. This bullet now enters a block of wood of mass 990 g. If the speed of the bullet is 500 ms^{-1}, find the speed of the gun and the common velocity of the wood.

Sol^n: mass of the bullet (m) = 10 g = $\frac{10}{1000} = 0.01$ kg

mass of the gun (M) = 1 kg

muzzle speed of the bullet (v) = 500 ms^{-2}

Speed of the gun (V) = ?

Using principle of conservation of linear momentum, we have

$mv = MV$

$\Rightarrow \quad 0.01 \times 500 = 1 \times V$

$\therefore \quad V = 5 \text{ ms}^{-1}.$

Again, combined mass of bullet and gun = 10g + 990g = 1 kg. Let V_1 be the common velocity of bullet and wood then momentum of the system before firing = momentum of the system of after firing,

$\Rightarrow \quad 0.01 \times 500 + 0.9 \times 0 = 1 \times V_1 \qquad \Rightarrow 5 + 0 = v \qquad \therefore v = 5 \text{ ms}^{-1}.$

9. A gun of mass 400 kg fires a shot of mass 3 kg, with a velocity of 200 ms^{-1}, find the constant force which acting on the gun would stop it after a recoil of 2.5 metres.

Sol^n: Now, momentum of the shot = mv = 3 × 200

momentum of the gun = MV = 400 × V

$\therefore$ Momentum of the shot = momentum of the gun

$\Rightarrow \quad 3 \times 200 = 400 \times V \qquad \Rightarrow V = \frac{3 \times 200}{400} = \frac{6}{4} \text{ ms}^{-1} = \frac{3}{2} \text{ ms}^{-1}$

Let, a be the retardation of recoil of gun. Then,

$0 = V^2 - 2as \qquad \Rightarrow V^2 = 2as$

$\Rightarrow \quad \left(\frac{3}{2}\right)^2 = 2 \times a \times 2.5 \qquad \Rightarrow \frac{9}{4} = 2 \times 2.5 \times a \qquad \therefore a = \frac{9}{4 \times 2 \times 2.5} = \frac{9}{20} \text{ ms}^{-2}$

Hence, the constant force acting on gun = F = ma = $400 \times \frac{9}{20} = 180$ N.

10. A shot of 2 kg is discharged by a gun of mass 400 kg with a velocity of 800 m/sec. Find the constant force which would be required to stop the recoil of the gun in (i) 2 metres, (ii) 1 $\frac{1}{4}$ sec.

Sol^n: mass of shot(m) = 2 kg

velocity of shot(v) = 800 m/s

mass of gun (M) = 400 kg

recoil velocity of gun (V) = ?

Now, mv = MV

$\Rightarrow$ $2 \times 800 = 400 \times V$ $\therefore$ V = 4 m/sec

Let F be the constant force required to stop the recoil of gun

(i) at distance 2 metres

Then $0^2 = V^2 - 2as$ $\Rightarrow 16 = 2a2$ $\Rightarrow a = 4\ m/s^2$

$\therefore$ $F = m \times a = 400 \times 4 = 1600$ N.

(ii) at time $1\frac{1}{4} = \frac{5}{4}$ seconds

Then $0 = V - at$ $\Rightarrow 4 = a.\frac{5}{4} \Rightarrow a = \frac{16}{5}$

$\therefore$ $F = m \times a = 400 \times \frac{16}{5} = 80 \times 16 = 1280$ N.

11. A gun of mass 1 metric ton, fires a shot of mass 14 kg and recoils up smooth inclined plane, rising a height of 1.6 m, find the initial velocity of the projectile.

Sol^n: Suppose u be the initial and v be the final velocity of the gun.

Now, $0^2 = u^2 - 2gh$

$\Rightarrow$ $u^2 = 2 \times 9.8 \times 1.6 = 31.36$

$\therefore$ $u = \sqrt{31.36} = 5.6$ m/sec

Again, we have

mass of the gun (M) = 1 metric ton = 1000 kg

velocity of the gun (V) = 5.6 m/sec

mass of the shot (m) = 14 kg

velocity of the shot (v) = ?

Now, MV = mv

$\Rightarrow$ $1000 \times 5.6 = 14 \times v$

$\therefore$ $v = \frac{1000 \times 5.6}{14} = 400$ m/sec.

12. A cricket ball of mass 150 g is moving with a velocity of 12 ms^{-1} and is hit by a bat so that the ball is turned back with a velocity of 20 ms^{-1}. The force of the blow acts for 0.01 s. Find the impulse and the average force exerted on the ball by the bat.

Sol^n: Here,

mass of the ball (m) = 150 g = 0.15 kg

initial velocity of the ball (u) = 12 ms^{-1}

final velocity of the ball (v) = 20 ms^{-1}

time duration (t) = 0.01 sec

Now, impulse of the force = change in momentum

= mass $\times$ change in velocity

$= 0.15 \times (120 - 12) = 0.15 \times 32$

= 4.8 MKS units of impulse.

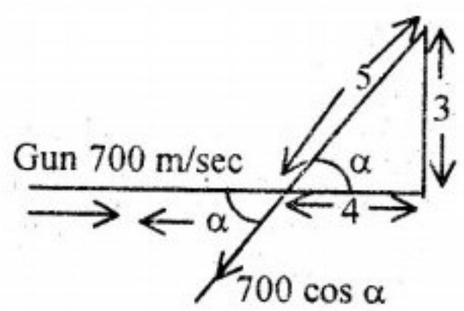
Again Let, R be the average force exerted on the ball by the bat

Then impulse of the force = R $\times$ t

$\Rightarrow$ $4.8 = R \times 0.01$ $\therefore$ $R = \frac{4.8}{0.01} = 480$ N.

13. A gun of mass 40 metric tones resting on an incline of 3 in 5, fires a shot of 100 kg horizontally with a velocity of 700 m/sec. Find the velocity of the recoil of the gun and the distance it moves up the incline before coming to rest. ($g = 9.8\ ms^{-2}$)

Sol^{n}: mass of the gun (M) = 40 metric tones = 40000 kg



Mass of shot (m) = 100 kg

Velocity of the shot (v) = 700 m/sec.

Velocity of gun (V) = ?

If α is the angle of inclination of the gun with the inclined plane.

Then from figure, $\sin\alpha = \frac{3}{5}$ and $\cos\alpha = \frac{4}{5}$

velocity of the shot in the direction of the inclined plane = 700 cos α m/sec.

Therefore its momentum = $100 \times 700 \times \frac{4}{5} = 56000$ ..... (ii)

Now, mv = MV

$\Rightarrow$ 40000V = 56000

$\Rightarrow$ $V = \frac{7}{5}\ ms^{-1} = 1.4\ ms^{-1}$.

Let s be the distance travelled by the gun along the horizontal, then acceleration due to gravity in the direction of horizon = -g sinα

Now, $v^2 = u^2 + 2as$

$\Rightarrow$ $0^2 = \left(\frac{7}{5}\right)^2 + 2(-g\sin\alpha)s$

$\Rightarrow$ $0 = \frac{49}{25} - 2 \times 9.8 \times \frac{3}{5} \times s$

$\Rightarrow$ $s = \frac{49 \times 5}{25 \times 2 \times 9.8 \times 3} = \frac{49}{294} = \frac{1}{6}$ m.

Hint and solution of MCQ's

1. Force is a vector quantity
2. When force acting on a body is zero, then momentum is constant.
3. The pull on a body of mass 5 kg due to the earth is Mg N = 5×9.8N = 49N
4. $Mg = 60 \Rightarrow m = \frac{60}{g} = \frac{60}{10} = 6$ kg
5. F = 20N

   m = 4kg

   $a = \frac{F}{m} = \frac{20}{4} = 5 m/s^2$
6. m = 1gm

   $a = 1\ cm/s^2$

   F = ma = 1dyne (the CGS unit of force is dyne)
7. The momentum of body = mass × velocity
8. m = 16kg, v = 20 m/s

   $\therefore$ momentum = mv = 320 kgm/s

9. $F = 5N, m = 0.2kg, t = 15$

$$F = \frac{mv - mu}{t}$$

$$\Rightarrow \quad 5 = \frac{0.2v - 0}{1} \quad \Rightarrow 0.2v = 5 \quad \Rightarrow v = 25m/s$$

10. Impulse = Force × time = Ft

11. $F = 15N, v = 18m/s, u = 6m/s, t = 4s$

$$F = \frac{mv - mu}{t} = m\left(\frac{v - u}{t}\right)$$

$$\Rightarrow \quad 15 = m\,\frac{12}{4} \qquad \Rightarrow m = 5\ kg$$

Impulse = Ft = 15 × 4 = 60Ns = 60kg m/s

12. $m = 50gm = 0.05kg$

$v = 20m/s$

$u = 0m/s$

$F = 2.5\ N$

$$\therefore \quad F = m\left(\frac{v - u}{t}\right)$$

$$\Rightarrow \quad 2.5 = \frac{0.05 \times 20}{t}$$

$$\Rightarrow \quad t = \frac{0.05 \times 20}{2.5} = 0.4\ sec$$

13. $m = 24\ kg, v = 15\ m/s, u = 0\ m/s, t = 5s$

$$\therefore \quad F = m\left(\frac{v - u}{t}\right) = 24 \times \frac{15}{5} = 72N$$

14. $F = 200\ kg$

$u = 50m/s$

$v = 0m/s$

$s = 25m$

Now, $v^2 = u^2 - 2as$ [a = average retardation]

$$\Rightarrow \quad 0^2 = 50^2 - 2a.25$$

$$\Rightarrow \quad a = \frac{50 \times 50}{50} = 50m/s^2$$

$$\therefore \quad F = ma = 200 \times 50 = 10{,}000\ N$$

15. $m = 49\ kg, F = 10kg, wt = 10 \times 9.8\ N, u = 10m/s, v = 0m/s$

Now, $-F = m\left(\frac{v - u}{t}\right)$ (Retarding force)

$$\Rightarrow \quad t = \frac{m(v - u)}{-F} = \frac{49 \times -10}{-9.8} = 5s$$

16. Force of body = mg N, acting vertically down,

Force of lift = ma, acting vertically up

$\therefore$ Pressure on lift moving up = ma + mg = m(a + g)

17. Force A body = 60g N, acting down

    Force of lift = 60N, acting down

∴ Pressure on lift moving down = mg – ma = 60g – 60 ×0.8 = 60 × 9.8 – 48 = 540N

18. m = 4.9 kg, u = 50m/s, v = 0m/s

    let a be the retardation for the body coming to rest then

    $v = u - at \Rightarrow 0 = 50 - a \times 5 \quad \Rightarrow a = 10 m/s^2$

    Then using Newton's law

    Resultant force, F = mg + ma = m(g + a) = 4.9 × 20 = 98N

19. m = 4.9 kg

    F = 12 tonnes wt. = 12 × 1000 × 9.8N

    Let a be the retardation then

    F = ma

    $\Rightarrow 12 \times 1000 \times 9.8 = 4.9 \times a \qquad \Rightarrow a = 24{,}000\ m/s^2$

20. m = 0.4kg (mass of bullet)

    M = 24 kg (mass of gun)

    v = 360 m/s (velocity of bullet)

    V = ? (recoil velocity of gun)

    using Newton's third law

    MV = mv

⇒ 24 × V = 0.4 × 360

⇒ V = 6 m/s

21. m = 0.5 kg, F = 2N, t = 15

    Change in momentum = F × t = 2 × 1 Ns = 2Ns

22. m = 0.05

    u = 200m/s

    v = 150m/s

∴ Impulse = m [v – (–u)] [being opposite]

= 0.05 [v + u]

= 0.05 × 380

= 19